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RESEARCH ARTICLE

SME crisis management and performance: leveraging algorithm supported induction to unravel complexity

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Abstract

This study contributes to the future directions of SME crisis management literature through algorithm supported induction by exploring the complex relationships between SMEs' strategic responses to the COVID-19 pandemic and their performance. Using data from the UK Longitudinal Small Business Survey, decision tree algorithms and explainable artificial intelligence techniques reveal how configurations of strategic actions and contextual factors shape performance outcomes. The analysis also uncovers dominant determinants and highlights previously overlooked non-linear and asymmetric relationships. Key findings emphasise the critical roles of responses to lockdown measures, utilization of the furlough scheme, and the interplay of firm size and age, which interact in complex configurations exhibiting asymmetry and non-linearity. This understanding provides a basis for informing future research directions, hypotheses, and strategies for SMEs to navigate crises and enhance resilience.

Keywords Crisis management · Algorithm supported induction · COVID-19 · SME performance · Machine learning · Configuration analysis · Complexity theory

1 Introduction

While the COVID-19 pandemic unleashed profound negative consequences across health, economic, and social landscapes [1, 2], the impact on individual firms was not uniform [3]. Businesses grappled with a myriad of unique challenges, ranging

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from staff layoffs, dwindling revenue, and supply chain disturbances to limited financial access, intermittent closures, and even outright failures [2–7].

Particularly susceptible were small and medium-sized enterprises (SMEs). Their inherent vulnerabilities, tied to their size, became more pronounced during the crisis—a phenomenon rooted in the 'liability of smallness' [7]. Further accentuating their plight were constraints such as restricted cash buffers [7–9] and their pronounced operations in sectors such as the service industry, which bore the brunt of pandemic-related restrictions, including social distancing measures [10].

Yet, the pandemic's disruptive force also paved the way for market shifts, both transient and enduring, alongside alterations in consumer behaviors [10–12]. This disruption unveiled novel business avenues [13, 14]. A subset of firms demonstrated agility in navigating these changes, seizing emergent opportunities, and consequently achieving success despite the adversity [3, 14–16]. This was particularly evident in areas such as online retail, e-learning, virtual meetings [10, 16, 17], medical equipment and pharmaceuticals [16], and select household goods [18]. Despite these patterns, there remains a gap in understanding why certain firms demonstrate superior performance amidst crises [19].

Firms can adopt a range of strategies when responding to crises, including pivoting to take advantage of opportunities, persevering with their existing strategy, and retrenching [20]. These strategies consist of specific actions, which varied in response to the crisis, with some firms relying on government support and others adapting existing resource bases and capabilities in response to changes in the market [21–23]. Firms that pivoted in response to the pandemic tended to perform better during and after the initial stages [21, 24, 25]. The emerging literature underscores the importance of dynamic capabilities for adaption [21, 22], but adapting also brings inherent risks [16].

Substantial government support was also available during the pandemic. Some studies advocate its importance in enabling firms to bolster productivity, invest, alleviate negative mental health impacts, increase earnings, preserve employment, and prolong their lifespan [26–28]. However, others have criticised the blanket support mechanisms for funding firms that were already performing poorly, for not allocating sufficient funding to regions where it is most needed, and for failing to consider the need for longer term recovery strategies [26, 29].

Although the importance of these internally and externally facilitated strategies and actions have been identified in past research [21, 25], less clarity exists on how firms integrate various actions into their overall strategic response and the subsequent performance implications. This is despite the considerable heterogeneity in firm responses to the pandemic, as acknowledged by Klyver and Nielsen [30], and theoretical arguments pointing to equifinality and substitutability in the underlying capabilities [31]. The effect of these strategic responses could also be impacted by contextual factors, such as firm size and age, further highlighting the potential for more complex relationships that require more research.

Reinforcing this view, the wider small business [32] and strategic management literature [33, 34], advocates for more research into configurational explorations. The lack of research on the configurations of strategic actions alongside contextual factors in aiding firms during pandemics leaves an uncharted territory with substantial potential. Navigating this space could not only enhance decision-making processes for SMEs but also sharpen the precision of government support initiatives and augment theoretical understandings of SME responses to crises. Configurational approaches assess various combinations of factors and pathways that can separately lead to the same or different outcomes [35]. They focus on more complex, non-linear and asymmetric relationships than conventional regression-based methods, which depend on linear dynamics and average net effects [35, 36]. The use of configurational approaches can therefore generate new insights into the combinations and complexity of mechanisms that shape outcomes, which would not be possible with traditional approaches [35, 36]. This represents a different way of conceptualising the research problem, compared with more traditional approaches.

The existing literature on SMEs responses to crises underscores the importance of strategic actions [37, 38] and government support [39, 40]. However, only nascent literature [25] highlights the joint role of strategic actions and government support in pandemic responses. In addition to the lack of research on more complex configurations of strategic actions and government support, there remains a knowledge void regarding the relative importance of specific strategic actions, representing another significant literature gap. The relative importance, or dominance, of predictor variables can have both theoretical and practical consequences [41]. In response to these gaps in the literature, this exploratory paper addresses the following question: How do configurations of strategic actions, interwoven with pertinent contextual elements, relate to SME performance in the backdrop of the COVID-19 pandemic?

The formulation of this question draws on complexity theory which brings to the fore the necessity to consider configurations of antecedents which may include equifinality and substitutability, alongside more complex non-linear and asymmetric relationships [35]. Whilst past work has drawn on techniques such as fuzzy set qualitative comparative analysis (fsQCA) to address complexity based questions [42–44], machine learning approaches are much less frequently used within this paradigm, but have shown potential in addressing these questions [45, 46]. Alongside identifying configurations of antecedents, machine learning methods also provide insight into dominant predictors, as well as identifying more complex non-linearity [46]. This allows us to go beyond both traditional regression based approaches, and more widely used configurational approaches such as fsQCA.

Although machine learning methodologies have begun to gain traction, they differ substantially from traditional regression based approaches both in the questions that can be addressed as well as the methods employed. This has resulted in the emergence of the analytics paradigm [47] within which we situate our study. Operating within this paradigm, we take an algorithm supported inductive approach to link our findings to theory, ensuring a meaningful theoretical contribution. Algorithm supported induction has been proposed as a means of leveraging advances in machine learning to inductively develop theory by identifying patterns in data [48]. This contrasts with the more traditional deductive hypothesis testing approach often adopted when using traditional regression based methods [46, 48], as well as contrasting with the traditional focus of machine learning to maximise predictive accuracy [49, 50], rather than inductive knowledge building. Although induction is normally situated within the purview of qualitative research, it can also play an important role in theory building from quantitative data [48]. We therefore adopt this approach to theory building through machine learning, rather than the more traditional hypothesis testing approach often adopted in quantitative studies.

Machine learning approaches are ideally suited to address our research question, facilitating the modelling of more complex heterogeneous relationships and offering measures of relative importance (dominance) [45]. Beyond the theoretical gaps previously mentioned, there have been explicit calls for a more robust integration of machine learning methodologies within entrepreneurship research [51, 52]. A limited body of literature within strategic management [33] and entrepreneurship [45, 46] has begun to incorporate machine learning methods. However, studies focusing on strategic crisis management in SMEs have not incorporated the use of machine learning to address this gap. Existing studies on SME strategic crisis management primarily applied standard regression analyses [27, 30], mainly focusing on linear relationships between variables.

Some researchers have attempted to capture more heterogeneous and non-linear relationships by utilising qualitative interviews to gather in-depth insights into firm behavior and decision-making during crises [53]. FsQCA has also been employed to analyse configurations of attributes enabling crisis management, allowing researchers to identify patterns across different firm profiles [43, 54, 55]. Additionally, structural equation modelling has been used to explore complex causal relationships, aiming to quantify the interactions and direct effects of entrepreneurial coopetive interactions and exogenous shocks, though its ability to capture non-linearity is still limited [56]. These approaches generally fall short in adequately modelling the evolving and multifaceted nature of crisis management, which is often characterized by complex and non-linear interactions that machine learning can more effectively address. The lack of studies incorporating machine learning methods creates the potential to gain additional insight by using machine learning approaches to capture more complex and heterogeneous relationships, alongside the identification of dominant predictors. This study addresses this gap in the literature by applying machine learning algorithms to understand the heterogeneity and dominance of firm responses to the COVID-19 pandemic, thus going beyond the existing literature.

The empirical study draws on data from the 2020 wave of the UK Longitudinal Small Business Survey. The analysis leverages two complementary tree-based machine learning algorithms: recursive partitioning and gradient boosting. Recursive partitioning yields an interpretable tree structure, facilitating the discernment of configurations of factors that influence performance shifts. Gradient boosting is a more complex algorithm that constructs an ensemble of trees, frequently demonstrating greater accuracy than individual trees [57–59]. By employing explainable artificial intelligence (XAI) techniques on the model generated by the gradient boosted machines (GBM), the findings show the relative importance of factors impacting changes in revenue, including their direction, shape, and magnitude of effect.

The results of the decision tree reveal three dominant overarching strategies, with each consisting of heterogeneous combinations of actions and firm characteristics.

The first strategy draws on seizing opportunities to grow, resulting in higher performance. The second involves a strategy of temporarily reducing operations and drawing on government support to survive. The third strategy is more heterogeneous in terms of the use of actions underpinned by dynamic capabilities and government support, also encompassing more complex interactions with firm size and age. SMEs adopting this strategy also had more mixed outcomes in terms of revenue performance. The findings suggest that SMEs' performance is closely associated with whether they drew on their capabilities to increase operations in response to lockdowns or relied on retrenchment enabled through the government's coronavirus job retention (furlough) scheme.

Beyond these factors, there is considerable heterogeneity in the configurations that lead to variations in performance, with contextual factors including firm age and size becoming important. The tree structure reveals asymmetry in these relationships, for example, with different configurations emerging for different sized firms. The black-box GBM is explored using XAI, revealing dominant predictors of increased operations due to the lockdown, use of the furlough scheme, and contextual factors of firm age and size. This helps to provide additional insight, as well as providing further validation of the single tree structure. Partial dependence plots (PDPs) applied to the GBM reveal non-linear patterns in these relationships.

The empirical evidence for complexity, enabled by algorithm supported induction, provides a basis for informing future research directions, hypotheses, and strategies for SMEs to navigate crises and enhance resilience, representing an advancement in crisis management and SME research. This study identifies configurations and heterogeneity in the strategic responses to COVID-19 and their impact on performance, alongside asymmetric and non-linear relationships. The findings reveal a spectrum of strategic manoeuvres, many of which leverage dynamic capabilities and governmental assistance, that SMEs adopted to navigate the COVID-19 pandemic and their consequent performance ramifications.

Furthermore, it identifies the dominant factors and strategic actions that resulted in different performance outcomes. These findings extend earlier relational studies by highlighting pivotal response mechanisms. This builds upon dominance analysis paradigms from management [60–63], and the broader social science literature [64, 65]. The methodological innovation of this study lies in its application of a machine learning configurational approach, situated within the analytics paradigm [47]. This extends existing literature [20, 66–68] that has predominantly centred on conventional linear methods with acknowledged shortcomings [36, 69]. Thus, the study contributes to the existing literature on strategic responses to crises [67, 70] and government support [39, 40].

The paper proceeds as follows. Section two reviews the main literature that informs this study, focusing on the configurations of strategic actions and government support. This is followed by the methodology in section three. The results and discussion of results are then presented in sections four and five, respectively, before the paper concludes.

2 Literature

2.1 Strategic responses to crises

In the wake of the COVID-19 pandemic, SMEs found themselves at a crossroads, faced with the critical task of navigating through unprecedented challenges while seizing emerging opportunities. The resilience and adaptability of these businesses were tested as never before, prompting a re-evaluation of strategies to ensure survival and growth. Building on the previous literature [20, 30, 53, 71], the focus of this section is on three strategic options that SMEs have adopted: pivoting or innovating, persevering with business as usual, and retrenchment [16, 20, 30, 53]. These overarching strategies consist of specific actions which firms can adopt in various configurations [30, 53].

2.1.1 Pivoting

Pivoting has emerged as an effective strategy for firms to adapt and innovate in response to the COVID-19 pandemic. It empowers SMEs to seize opportunities to expand operations, offer new products or services and engage in new ways of doing business. Emerging evidence shows that some firms adopted a pivoting strategy in response to the COVID-19 pandemic [18, 20, 23, 72], by introducing business model changes [20, 23, 72], offering new products or services [16], mobilizing slack [73], and adopting innovation based strategies [30], leading to improvements in performance [74]. However, pivoting carries significant risks and can, in certain circumstances, diminish performance or even result in failure [16].

One of the ways to gain a deeper insight into the foundations of pivoting strategies is to explore dynamic capabilities theory. This theory illuminates the processes through which firms can methodically adapt and reconfigure their operational resources and capabilities to respond to swift and significant market shifts, enabling firms to take advantage of opportunities and better meet the needs of customers [75, 76]. Dynamic capabilities theory also posits that SMEs without developed dynamic capabilities might instigate change via “ad hoc problem solving” [77], which then departs from routinely seeking change through which dynamic capabilities ordinarily develop [78] and thus could potentially be more dependent on the characteristics of the organization. Dynamic capabilities are pivotal to a firm’s ability to adapt organizational capabilities and resources in general [79, 80], but especially in response to crises [21, 23, 81–83]. Their effective utilization could enable some companies to benefit from crises [84–86].

Research focused on strategic responses during COVID-19 found mainly positive but varied outcomes on firms’ responses [2, 14, 21, 87, 88] but neither mechanisms of dynamic capabilities [89] nor their effect on performance [90–92] are fully understood. The performance impact is likely to be contingent on the type of dynamic capability and the relevance of that specific capability’s recurrent function in a given context [93].

2.1.2 Perseverance

An alternative response to crises for firms is the perseverance strategy, which involves continuing with business as usual [20]. In some cases, this strategy can lead to better performance compared to strategies involving adaptations [94]. However, implementing a perseverance strategy requires having slack resources to absorb the impact of the crisis [20], which can be challenging for SMEs due to a lack of funding during such times [7, 95]. In these cases, debt financing could be a viable option [72]. This strategy also carries the inherent risk of being effective only in the short to medium term, possibly jeopardizing the firm's long-term viability [72].

Much of the COVID-19 assistance was focused on sustaining firms and, in this way, potentially directed many SMEs towards perseverance. The rationale for these schemes predominantly revolved around preserving SME continuity rather than fuelling innovation. For instance, the then Chancellor of the Exchequer, Rishi Sunak, in his economic response to COVID-19, advised using support measures to manage expenses such as paying "rent, the salaries, suppliers, or purchase stock." [96]. The most notable schemes included the HMRC time to pay scheme, VAT deferral, business rates holiday and related loans. Evidence from the US points to firms using COVID-19 loans to make non-payroll fixed payments and build savings buffers [97]. These support mechanisms were designed to delay the business's costs, thereby enabling the business to continue operating during the pandemic. In contrast to the furlough scheme, they did not imply a reduction in the firms' asset base or operations, and in some cases, only delayed payments rather than permanently reducing the firms' cost base.

Although these SMEs can maintain a level of continuity and "business as usual", it is unlikely that they will experience improvements in performance. The reliance on a cash influx from the government is likely to have a negative impact on some SMEs because they could potentially lose their long term competitive advantage to firms that further developed their capabilities through experience gained during the COVID-19 pandemic, of which there are already numerous examples [13, 30, 98]. Furthermore, some firms may not be viable and enabling these distressed firms to stay afloat could potentially result in negative, longer-term consequences through a lack of adjustment in the aggregate production capacity after a negative demand shock [99, 100] with a suggestion that enabling less viable firms could "transform the temporary shock due to COVID-19 into a permanent shock by distorting the liquidity supply toward inefficient firms" [101].

2.1.3 Retrenchment

Retrenchment is a strategy that involves cost cutting and reducing assets, utilizing measures such as downsizing and reducing staff or narrowing the business's scope by retracting products or divesting assets [20, 102, 103]. A retrenchment strategy is commonly used as a first step in turnaround, with the aim of survival and increasing cash flow [103], which can be followed by a recovery stage focused on increasing growth [103].

However, the adoption of a retrenchment strategy can negatively impact performance by creating adjustment costs and destroying synergies between the continuing business and the reduced elements [102, 104]. For example, focusing on declining firms in the software industry, Ndofor et al. [102] find that adoption of a retrenchment strategy is negatively related to performance, whereas more growth orientated strategies such as: “new product development, strategic alliances, and acquisitions” were positively related to performance. Furthermore, there is some debate about whether retrenchment should be adopted as part of a longer-term turnaround strategy since it is often quickly followed by restructuring efforts with less emphasis on cash flow [105].

Retrenchment strategies were commonly used by SMEs during the COVID-19 pandemic [106], as evidenced by the staffing reductions implemented by many firms [6, 107]. In the UK, the furlough scheme helped to mitigate staffing reductions by providing support to cover up to 80% of employees’ salaries and other employment costs for staff who were temporarily not working in the business [27]. This scheme helped to safeguard jobs that may otherwise have been lost due to redundancy, with around 1.16 million jobs on furlough and an estimated cost of £70 billion at the end of the scheme [108]. The use of the furlough scheme could therefore provide an indication of the firm’s retrenchment efforts in seeking to temporarily reduce staffing costs [53].

2.2 Embracing complexity and configurational diversity in SMEs’ strategic actions

Although the literature distinguishes between strategic responses to the pandemic and the potential for different outcomes from these responses [20], firms have a diverse range of specific actions that can be taken as part of these strategies. These mechanisms can be used in different combinations, potentially leading to different outcomes. Additionally, equifinality can occur, where different pathways result in the same outcome due to varying combinations of mechanisms [31, 109]. This creates the potential for heterogeneity in the actions that firms take as part of their strategy.

The confluence of strategic responses, government support, and contextual factors signifies more than broad strategies; rather, it represents configurations shaped by a range of internal and external determinants. Some strategic responses, such as pivoting, relied on actions underpinned by dynamic capabilities [110] that interact in various ways and may substitute or complement each other depending on the conditions [31, 111] and thus result in different performance outcomes [18, 43, 44].

Given the unprecedented complexities introduced by the COVID-19 pandemic, there arises a pressing need to re-evaluate traditional methodologies that probe strategic responses of SMEs. Historically anchored linear models have shaped much of the understanding, often isolating strategic actions or offering oversimplified paradigms [36]. Yet, the complex challenges manifested by the pandemic have spotlighted the inadequacies of linear perspectives [30], necessitating a sophisticated approach attuned to the dynamic interplay of multifarious factors.

Within this context, complexity theory emerges as a particularly salient framework, challenging the 'one size fits all' approach. Fundamentally premised on the notion that systems are more than sum of their parts, this theory highlights the implications of more complex interdependencies, which can result in unexpected outcomes [112]. This paradigm is particularly pertinent in the context of the pandemic, where firms were forced to navigate a complex mix of challenges, devising strategies that are shaped by an amalgamation of their capabilities, governmental interventions, and overarching contextual nuances.

Fostering this perspective, configurational approaches, deeply entrenched in complexity theory, champion the exploration of heterogeneous and non-linear associations, markedly deviating from conventional regression-centric methodologies [36]. Rather than focusing on linear relationships, configurational approaches focus on the combinations of factors that occur together, leading to one or more outcomes [69, 113, 114].

The growing emphasis on configurational approaches in the management literature is indeed an acknowledgment of this complexity. While methodologies like fsQCA have been pivotal in elucidating these configurations [91, 114–117], the integration of machine learning, especially decision trees, offers further refinement and extension [33, 45, 118]. The inherent design of decision trees, which parse complex decisions into a cascade of binary choices predicated on influential factors, is congruent with a configurational outlook. Moreover, their algorithmic nature enables the assimilation and processing of large datasets, potentially revealing insights that might otherwise remain obscured. In contrast to the machine learning approach, more widely applied traditional regression based approaches [27, 30] are limited in their ability to identify more complex configurations, non-linearity, and asymmetric relationships. Although techniques such as fsQCA can be used to identify configurations, the machine learning approach can also be used to gain additional insight about predictor dominance and non-linearity, in addition to identifying complex configurations and asymmetric relationships [46]. Techniques such as partial dependence plots also provide insight into the shape of relationships in the data, allowing for the study of more complex non-linearity and floor and ceiling effects [46]. These advantages of machine learning are important in addressing our overarching research question, as well as addressing wider calls for the increased use of machine learning in entrepreneurship research [45, 51].

Previous crises have shown that firms can draw on a variety of configurations of capabilities and cognitive factors as part of their adaptation strategies [34]. Contextual factors also play a crucial role, with varying reliance on dynamic capabilities among different firms and industries [82, 116]. In addition, firms used a combination of government support and dynamic capabilities to adapt to the pandemic, resulting in different performance outcomes [21, 27]. Government support has also been found to be important in configurations in contexts such as entrepreneurship [119, 120], business exit [115], survival of green companies [121], and the development of dynamic capabilities [122].

3 Methodology

To address the research question we draw on algorithm supported induction [48], which we view as situated within the emerging business analytics paradigm [46, 47]. This approach focuses on the use of machine learning algorithms to inductively build theory through the identification of patterns in data [48]. These theories can then be tested in future studies using different samples and more traditional hypothesis testing approaches [48, 123].

Beyond considerations of theory building and testing, this approach also differs from the traditional hypothesis testing approach in terms of how models are evaluated and interpreted [46]. Models built using machine learning are typically evaluated based on predictive accuracy on a separate test dataset, rather than focusing on model fit and analysis of residuals [124]. Machine learning models are usually interpreted differently, with more focus on the relative importance of predictors, interactions, and non-linearity, compared with traditional regression based analyses which emphasise statistical significance and effect size [46, 123]. Another benefit of the machine learning approach, which is particularly relevant to our study is that it also enables more information to be analysed as individual variables do not need to be combined into constructs [125]. This allows us to include a range of granular measures, distinguishing their unique contributions.

Other widely applied methods, such as regression and fsQCA, are less well suited for algorithm-supported induction compared with machine learning methods [46, 48]. Regression typically requires a pre-specified model and focuses on testing hypothesized relationships, which limits its capacity to uncover unexpected patterns in an inductive, data-driven manner. Similarly, while fsQCA is effective in identifying configurational patterns, it often depends on researcher-calibrated thresholds and unlike machine learning approaches, does not inherently facilitate the exploration of vast data landscapes.

Machine learning algorithms are ideally suited for algorithm supported induction, enabling the identification of more complex patterns than can be detected using traditional regression based approaches [123]. The machine learning algorithms allow patterns to be identified 'that would not always be intuitive for humans to identify but intuitive enough to understand once found' [48]. These patterns are learned directly from the data rather than being specified in advance [123]. The ability to model more complex relationships also often results in more accurate models, with several studies highlighting improvements in accuracy compared with more traditional regression based approaches [46, 126]. Although machine learning models focus on identifying patterns, or associations, in data they can also be used to identify causal relationships [48].

This approach enables us to focus on more complex configurations and interactions amongst variables. Following the model building, the next stage in this process is to provide explanations for the observed patterns [48], which we focus on in our discussion section. This approach is similar to the abductive approach to theory building from supervised machine learning proposed by Chou et al. [126], in which theory and explanations are derived from patterns observed from machine learning models. This echoes calls in the wider management research for more ‘empirical discovering’ [127]. The approach to building theory from patterns identified by the machine learning algorithm is similar to the process of theory building often undertaken in qualitative research [126].

3.1 Data

The study draws on data from the 2020 wave of the Longitudinal Small Business Survey [128]. The survey samples small and medium sized businesses with less than 250 employees in the UK, focusing on various aspects of the businesses as well as their performance. Data for the study was collected via a telephone survey carried out between September 2020 and April 2021. Data was therefore collected after the initial stages of the pandemic. This timeframe provided businesses with an opportunity to implement response mechanisms and to utilise government support. It also allowed for an assessment of the relationship with performance.

3.2 Dependent variable

The dependent variable in this study is the firm’s change in revenue turnover over the past year. Firms were asked to respond to the following question: “Compared with the previous 12 months, has your turnover in the past 12 months increased, decreased or stayed roughly the same?”. Responses to this question therefore result in an ordinal variable, which can take three values indicating whether revenue increased, stayed the same, or decreased over the past year. This variable provides an indication of SME performance during the early stages of the COVID-19 pandemic, comparing revenue levels before the pandemic with those after the initial stages. Similar measures of performance have been used in related studies [21, 25]. 186 observations had missing data on the dependent variable and were therefore excluded from the analysis.

3.3 Independent variables

The independent variables in this study focus on strategic responses, government support, and contextual factors. In line with previous scholars [21, 79, 129, 130], this study utilises study-specific measures to evaluate strategic actions rooted in dynamic capabilities that are relevant to this investigation, capturing the specific sub-channels of interest.

To facilitate more in-depth configurational approaches, these actions are quantified using a series of dummy variables that take a value of 1 when SMEs adapted to COVID-19 by increasing operations, changing products or services, changing processes, or changing methods of selling. With regards to increasing operations, respondents were asked: “Which of the following statements best describes how your business adapted during the lockdown restrictions?” This was coded 1 where the respondent reported “Operations were increased” else 0. This question captures the SMEs’ specific response to lockdowns, rather than more general changes in operations. To capture the latter three elements, respondents were asked: “Which of the following measures has your business taken to mitigate the impacts of the pandemic and any associated trading restrictions?” Responses were coded as 1 for answers of “yes” to “Changed services/products provided”; “Changed processes/ways of working”; and “Changed methods of selling”.

A series of dummy variables are also included to capture whether SMEs received support from government assistance programs. These variables are constructed based on the responses to questions on whether firms used COVID-19 support mechanisms, including HMRC time to pay, VAT deferral, business rates holiday, self-employed income support or furlough scheme. For each scheme respondents were asked “Has your business received funds from?” followed by the specific scheme.

In addition, data related to context and firm characteristics are included. The contextual variables encompass the following: size of the workforce, comprising employees, owners, and temporary staff; the proportion of owners who are female; whether the business is family-owned; year of establishment; and industry sector. The inclusion of these additional variables enables the identification of how both firm and contextual factors interact with the internal and external response mechanisms.

Considering contextual factors is crucial in configurational approaches [69], as indicated by existing literature. For example, the significance of contextual factors such as sector is emphasized by Miklian and Hoelscher [95], whilst Cesaroni et al. [131], Li et al. [132], and Clemente-Almendros et al. [133] highlight the importance of gender in response to crises. The full list of variables and descriptive statistics are presented in Table 1.

3.4 Analysis

The analysis follows a typical machine learning process, which involves data splitting, model training and tuning, model evaluation and interpretation [134]. The full dataset is first split into training and testing datasets. Random stratified sampling based on the dependent variable is used to ensure that the distribution of the data across the dependent variable is maintained. The first dataset consists of 80% of the observations, and is used to train the models. The remaining 20% of the observations form the second dataset that is used to test the models. Importantly, the test data is not involved in the training process at all, providing a more objective way of assessing the model’s performance [134].

Two tree based algorithms, recursive partitioning and gradient boosted machines, are employed to model the relationships between the independent and dependent variables. These algorithms are selected because they are complementary, offering different insights about the data. The decision tree is particularly useful for identifying configurations [45], whereas gradient boosting has been found to be particularly accurate [57, 59]. Both algorithms have been successfully used in past research in the business and management literature [33, 45, 59, 135–138].

Table 1 Descriptive statistics and variable descriptions

Variable	Description	N	Mean	SD	Min	Max
turnover_change.up	Increase in turnover	7433	0.18	0.38	0	1
turnover_change.same	Maintained the same turnover	7433	0.26	0.44	0	1
turnover_change.down	Decreased turnover	7433	0.56	0.5	0	1
cov_ops.yes	Adapted to lockdown restrictions by increasing operations	7433	0.07	0.25	0	1
cov_new_prod.yes	Mitigated the impact of the pandemic by introducing new products/services	7433	0.21	0.41	0	1
cov_new_work.yes	Mitigated the impact of the pandemic by introducing new ways of working	7433	0.62	0.48	0	1
cov_selling.yes	Mitigated the impact of the pandemic by introducing new ways of selling	7433	0.27	0.44	0	1
HMRC_time_to_pay.yes	Used HMRC time to pay scheme	7433	0.09	0.29	0	1
deferred_VAT.yes	Used deferred VAT scheme	7433	0.27	0.44	0	1
rates_holiday.yes	Used rates holiday scheme	7433	0.17	0.38	0	1
self_emp_inc_support.yes	Used self employment income support scheme	7433	0.12	0.32	0	1
furlough.yes	Used furlough scheme	7433	0.53	0.5	0	1
no_employees.1	No employees	7433	0.27	0.44	0	1
no_employees.2	0–9 employees	7433	0.36	0.48	0	1
no_employees.3	10–49 employees	7433	0.25	0.43	0	1
no_employees.4	50–249 employees	7433	0.12	0.32	0	1
no_owners	Number of working owners	7075	1.92	1.76	0	32
no_temps	Number of temporary staff	7433	1.48	8.16	0	180
sector.1	Production and construction	7433	0.24	0.43	0	1
sector.2	Transport, retail and food service/ accommodation	7433	0.27	0.44	0	1
sector.3	Business services	7433	0.34	0.47	0	1
sector.4	Other services	7433	0.15	0.36	0	1
bus_age	Year founded	6793	1995	23.23	1400	2019
fam_bus.1	Family led business (yes)	7433	0.76	0.42	0	1
fam_bus.2	Family led business (no)	7433	0.23	0.42	0	1

Table 1 (continued)

Variable	Description	N	Mean	SD	Min	Max
fam_bus.3	Family led business (refused)	7433	0.003	0.05	0	1
women_own.1	Women led	7433	0.17	0.38	0	1
women_own.2	Equally led	7433	0.2	0.4	0	1
women_own.3	Women in minority	7433	0.12	0.32	0	1
women_own.4	Entirely male led	7433	0.46	0.5	0	1
women_own.5	Don’t know	7433	0.05	0.21	0	1

Recursive partitioning builds a model by iteratively splitting the data into increasingly homogenous groups based on the target variable. The algorithm starts with the full dataset and tests each variable to find the variable that results in the most homogenous groups when the data is split into two parts based on that variable. This splitting process is repeated for each subsequent group, resulting in sub-groups that are increasingly homogeneous in terms of firms which decreased, maintained, or increased revenue. This process of recursive partitioning continues until a stopping criterion is reached, which in this case is a maximum complexity parameter, which restricts the size of the tree by only splitting on variables that do not decrease the accuracy by a factor of the complexity parameter.

Gradient boosted machines are also a tree based algorithm, but unlike recursive partitioning, which uses a single tree, the algorithm builds an ensemble of trees. The algorithm starts by constructing and evaluating a single tree, and then builds subsequent trees, with each subsequent tree aiming to correct the error in the previous tree. Typically, hundreds or thousands of trees are built to generate a single model. Each decision tree is usually a weak learner, in that it has a limited depth. This approach often results in more accurate models than a single decision tree [57–59]. An advantage of tree based algorithms is that they can deal with missing values by including these in the tree structure or using surrogate splits. This means that nearly all observations could be included in the analysis, resulting in a final sample size of 7433.

The parameters of both the single tree and GBM algorithms are tuned using five fold cross validation, repeated ten times. This approach splits the training dataset into five equal parts, and uses four of these parts to build the model, whilst testing the performance on the fifth part. This is repeated for each tuning parameter, and is further repeated ten times. The aim of this process is to identify the tuning parameters that result in the most accurate model. After this process has been completed, the accuracy of the model is evaluated by making predictions on the test dataset. The overall model accuracy is evaluated primarily using the multiclass area under the curve of the receiver operating characteristic (AUC-ROC), alongside kappa, sensitivity, specificity, and balanced accuracy.

Recursive partitioning has the advantage of generating a series of interpretable rules which can be visualised in a tree consisting of branches and nodes. This allows for the study of different configurations of factors that lead to outcomes of interest [33]. However, the gradient boosted machine is more difficult to interpret as multiple decision trees have been combined. Developments in XAI, such as permutation variable importance and partial dependence plots (PDPs), can improve the interpretability of the tree. Permutation variable importance allows us to determine the relative importance of the independent variables by permuting each variable in turn and evaluating the decrease in accuracy that follows [139].

PDPs enable the evaluation of the direction, shape, and strength of the relationship between each independent variable and the dependent variable. The PDP algorithm calculates the average predicted value for each unique value of a particular variable. This is accomplished by setting all observations of a focal variable to be a particular unique value, and then using the model to make predictions, which are then averaged [140]. This process is repeated for each unique value of the focal variable, providing a measure of the impact of each value of a particular independent variable on the dependent variable [140]. Since most of the independent variables are binary, the PDPs show the probability of the outcome (revenue increasing), when the variable takes a value of 0 or "no", and when it takes a value of 1 or "yes". For continuous predictors, PDPs also allow non-linear relationships to be evaluated, as no linear structure is imposed on the data.

4 Results

4.1 Results from the decision tree configuration

The decision tree depicted in Fig. 1 elucidates the configurations of strategic actions, contextual factors, and SME performance outcomes, presenting a nuanced representation of these relationships. The tree first splits based on whether the firm increased operations in response to the COVID-19 lockdowns. Firms that increased operations in response to lockdowns are predicted to increase revenue during COVID-19, accounting for 7% of the observations. Among the firms that did not increase operations in response to lockdowns, the tree subsequently splits based on whether the firm used the furlough scheme. Firms that did not increase operations but did use the furlough scheme are associated with decreased revenue during COVID-19, which comprise 51% of the observations.

SMEs that neither increased operations in response to lockdowns nor used the furlough scheme exhibit more heterogeneity in the factors that influence performance. Within this group of SMEs, micro-businesses and businesses with no employees are predicted to have decreased revenue, accounting for 25% of total

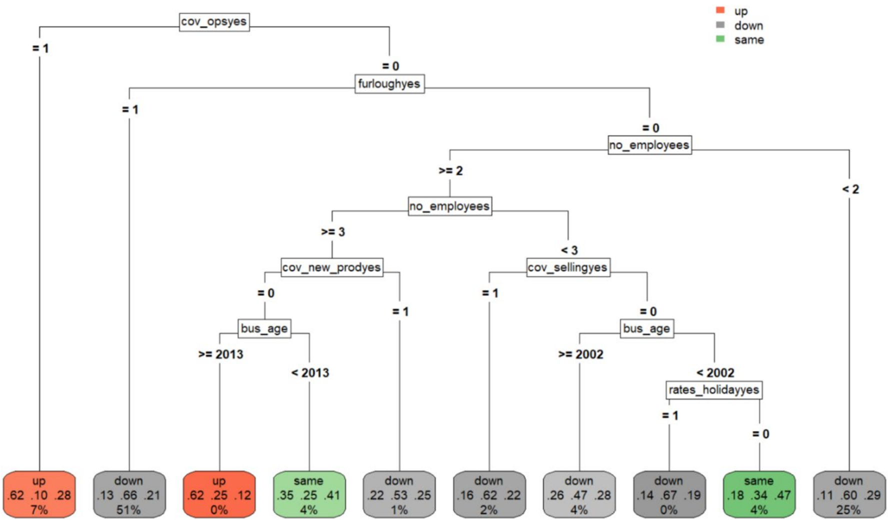

Fig. 1 Decision tree model showing the configurations of factors associated with performance outcomes

observations. For larger SMEs, two configurations result in no change in revenue, while five configurations lead to a decrease. Larger SMEs that did not introduce a new product, and are older are predicted to have no change in revenue. Similarly, SMEs with 1–9 employees, who did not change selling methods and are older and did not benefit from the rates holiday are also predicted to have no change in revenue.

4.2 Results of the dominance analysis and partial dependence plots

The dominance analysis and PDPs utilise the model produced by the GBM, notable for its refinement in discerning complex relationships between strategic actions and SME performance. The value of the GBM model, and by extension the dominance analysis, lies in its ability to sort and highlight variables based on their contribution to understanding the dynamics at play. Figure 2 shows the variable importance for the sixteen variables included in the study. The most important predictor of the change in revenue is whether the firm increased operations in response to the COVID-19 lockdowns, followed by whether the firm used the government support furlough scheme. After these two variables, the next three most important variables are firm characteristics, including the number of employees, the age of the business, and the sector it operates in. This is followed by the number of owners, the gender of the owners, and whether the SME introduced new products/processes, or new ways of selling to mitigate the effect of the pandemic.

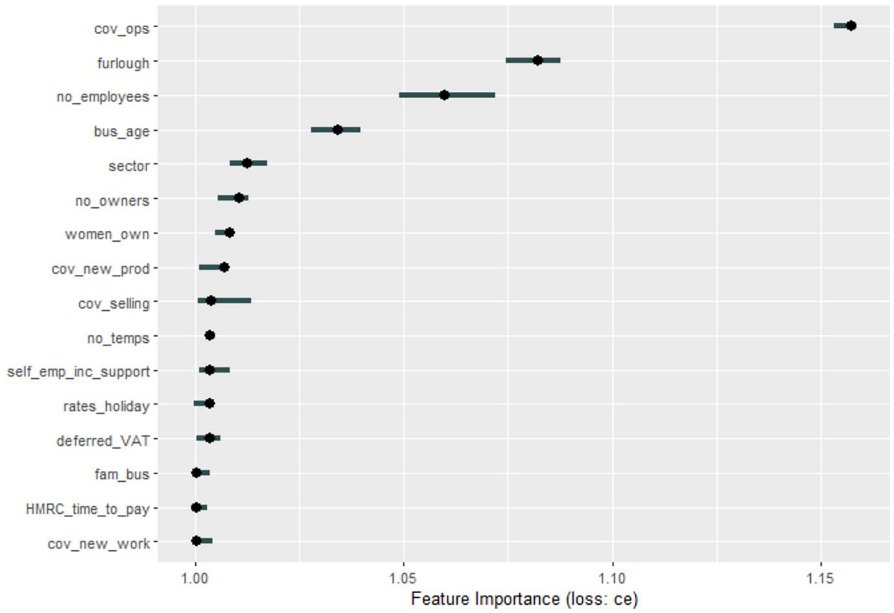

Fig. 2 Permutation variable importance for the independent variables

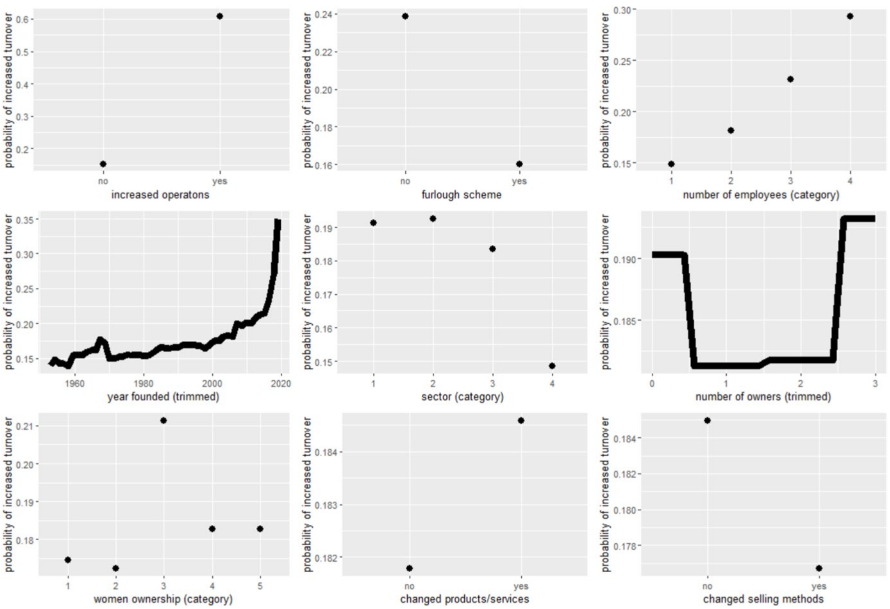

Fig. 3 Partial dependence plots for the nine most important variables

PDPs are used to examine the relationship between the most important predictors of increased revenue during COVID-19, based on the GBM model. Figure 3 shows the plots for the top nine predictors. Firms that increased operations in response to lockdowns had a higher probability of increasing revenue, with a probability of 0.61 compared to 0.15 for those that did not. Conversely, firms that used the furlough scheme had a lower probability of increasing revenue (0.16) than those that did not (0.24). Larger SMEs and newly founded firms were more likely to increase revenue. SMEs in the production and construction sectors were more likely to increase turnover, while those in other services had the lowest probability (0.15) of increasing revenue. SMEs with no working owners and those with three working owners have a slightly higher probability of increasing revenue. SMEs with a majority of male owners and those with only male owners had a slightly higher probability of increasing revenue. Changing products or services to mitigate the effect of the pandemic slightly increased the probability of increased revenue for SMEs, but changing selling methods had a less positive impact.

4.3 Model assessment

Although prediction is not the focus of the study, the overall performance of both models is evaluated using a separate holdout test dataset. The decision tree exhibits moderate discriminative ability with a multiclass AUC-ROC of 0.61. The

GBM shows a slight uptick in discriminative ability with an AUC-ROC of 0.66. Additional accuracy metrics are presented in Appendix 1.

5 Discussion

This study draws on algorithm supported induction to extend existing theory on SME crisis management by highlighting more complex heterogeneity, non-linearity, asymmetry, and dominance than considered in past studies. This is facilitated by the use of tree based machine learning algorithms, including a single decision tree, and an ensemble of trees (GBM) which is interpreted using approaches from XAI including permutation variable importance and PDPs. The decision tree allows us to examine configurations of factors related to the performance of SMEs during the pandemic, whilst the XAI approaches allow us to probe the black box GBM, revealing dominant factors and the shape of their relationship with performance. In this discussion we synthesise the key findings from both the single decision tree and the GBM, illustrating the key contributions from the study with relation to our research question. The findings are related back to the wider crisis strategy literature, which posits crisis response strategies of pivoting, perseverance, and retrenchment [16, 20, 30, 53].

Central to our exploration is the interplay between government support, strategic actions, and contextual factors that collectively shape the performance of SMEs' in navigating the challenges presented by the pandemic. The nuanced exploration of contextual factors, such as firm size, age, and gender leadership, provides deeper insights into the differential impacts of the pandemic, suggesting that performance during the pandemic was contingent upon these characteristics, which interact with strategic actions. This extends the theoretical discourse [16, 20, 30, 53] by illustrating the complexity and heterogeneity of strategic adaptations and contextual factors in crisis situations.

By integrating complexity theory with machine learning techniques, this study unveils the nuanced ways SMEs navigate crises. This dual approach not only augments existing theoretical models [20, 21, 67, 141], but also introduces a novel methodological paradigm for analysing strategic decisions in crisis settings. This study distinguishes itself through the application of complexity theory via machine learning, illuminating the nuanced strategies employed by SMEs during this tumultuous period, along with their effectiveness. Additionally, the findings highlight the dominant strategic actions and contextual factors, further extending the emerging domain of management research centred around dominance analysis [60, 142, 143]. To address the research question, the core theoretical contributions are discussed in more detail below.

Informed by complexity theory, the exploration in this study underscores the heterogeneous and non-linear dynamics that define strategic decision-making under unprecedented circumstances such as the COVID-19 pandemic. The interaction of diverse components—be they strategic actions facilitated by internal capabilities or external support mechanisms—can lead to emergent, often unpredictable outcomes for SMEs. Three distinct configurations are identified from the single decision tree based on whether SMEs leveraged their capabilities to increase operations in response to lockdowns, or adapted by introducing new products or methods of selling, or whether they utilised the furlough scheme. Configurations are further distinguished based on firm size and business age. These configurations are interpreted in the context of the literature on strategic responses to crises, with a focus on persevering, pivoting, and retrenchment strategies [20].

The first group of SMEs observed in the tree structure are those that increased operations in response to COVID-19 lockdowns. These firms performed relatively well during the pandemic and were able to increase revenue despite the challenges. The results of the variable importance from the GBM further illustrate the importance of adapting by increasing operations, with this being the most important predictor of performance, whilst the PDP shows the positive relationship with performance. Such firms may have adopted a pivot strategy in response to the pandemic, taking advantage of the opportunities that emerged, for instance, by innovating and diversifying their offerings with new products and services [2, 44]. To execute such a pivoting strategy, firms require the ability to increase operations and take advantage of opportunities, pointing towards the importance of dynamic capabilities in enabling firms to adapt to the challenges and opportunities posed by crises. However, the majority of SMEs did not fare as well during the pandemic, as illustrated by the second and third configurations.

The second group of SMEs identified by the tree structure are those that did not respond to lockdowns by increasing operations, instead opting to use the government furlough scheme. This response is in line with a retrenchment strategy, which involves "a strong emphasis by the firm on cost and asset reductions as a means to mitigate the conditions responsible for financial downturn" [103]. In this strategic response, SMEs temporarily reduced their staff numbers to decrease operational costs and relied on government support. As a result, they were more likely to experience a decrease in revenue.

The PDP also supports the negative association between the use of the furlough scheme and revenue performance observed in the decision tree. The variable importance measures highlight the overall importance of use of the furlough scheme as the second most important determinant of performance. The lower level of performance of these SMEs may be an indication that the SMEs relying on external support through the furlough scheme may be the least equipped to deal with the external shock. It is likely that some of the SMEs that availed of the furlough scheme were already struggling and required external assistance to cope with the crisis, while stronger firms could rely on their existing capabilities to adapt. Additionally, Belghitar et al. [26] contend that government support schemes often did not benefit the firms that most needed the support.

The retrenchment strategy requires government support mechanisms to enable firms to temporarily reduce operations and staffing numbers. Indeed, government support mechanisms, such as the furlough scheme, may have encouraged firms to engage in retrenchment strategies by temporarily reducing staffing levels, thereby reducing their potential to increase revenue during the pandemic. On the other hand, without the government support mechanisms, these firms would be at higher risk of staff redundancies or failure [26].

The findings contribute to the literature on strategic responses to COVID-19 by showing that retrenchment is associated with poorer performance, suggesting that findings from past studies on other crises [102, 104] are also applicable during COVID-19. For example, in the context of a declining software industry, Ndofor et al. [102] find that firms performed better when they took strategic actions, such as introducing new products, rather than retrenchment. However, Jibril et al. [27] identify a more positive impact of government support, finding that firms who avail of furlough and loan schemes are more likely to plan investments.

The third configuration observed in the decision tree includes SMEs that did not increase operations due to lockdowns and did not use the government furlough scheme, instead focusing on the introduction of new products or services, and new ways of selling. By not increasing operations and not using the furlough scheme to temporarily reduce staffing, firms in this configuration may be engaging in a perseverance strategy, with some subsets of firms adopting pivoting actions alongside perseverance. This group represents a more diverse set of firms, and their performance outcomes depend heavily on contextual factors including the firm's size and age. Although the existing literature notes the importance of contextual factors [21, 144], our results extend beyond this literature in highlighting the unique configurations of factors that are related to performance across subsets of SMEs. This provides more granular insights about the characteristics of subsets of firms and their associated performance outcomes than could be obtained from traditional regression based approaches, for example, by including moderating variables.

The results of the PDPs further elaborate on the role of contextual determinants, demonstrating that older firms were less likely to increase their performance, whereas larger firms were slightly more likely to do so. The findings presented in this study support past studies that have highlighted the importance of age in firm responses to crises, with younger firms exhibiting higher growth [145] and survival rates [146] during the 2008 financial crisis. Newer firms also demonstrated increased investment following the crisis [147] and were more likely to adapt to the COVID-19 pandemic [2].

The PDP shows that businesses led entirely or in the majority by males are only very slightly more likely to increase revenue, compared with women led businesses, suggesting that gender diversity in the leadership of SMEs plays a less substantial role than government support and dynamic capabilities. The results therefore highlight a more nuanced relationship, where an effect is significant, but the impact of the variable on prediction is small. This corresponds with the mixed findings from the wider literature, with some studies finding that the pandemic has had a disproportionately negative impact on women-owned businesses [144], with others finding a positive impact of female managers on innovation [133], as well as studies which fail to identify gender differences [2, 27].

The overall structure of the tree highlights asymmetric relationships, not previously discussed in the literature. For example, there are differences in the relationship between increasing operations in response to the lockdown, compared with not increasing operations, with the latter relationship depending on configurations of factors including firm size and age. In contrast, increasing operations in response to the lockdown directly and positively impacts performance. The tree structure also shows that firms who use the furlough scheme generally experience a decrease in performance, whereas those who do not have much more nuanced outcomes contingent on contextual factors. This asymmetry is a key tenant of complexity theory, but is examined much less frequently than the symmetric relationships studied by traditional regression based approaches [35].

The study also contributes to dynamic capabilities literature [21, 23, 81–83], highlighting the importance of considering more complex configurations, non-linearity, and the relative importance of strategic actions underpinned by dynamic capabilities. Dynamic capabilities which underpin the firm's ability to increase operations in response to external turbulence are important for increasing performance during a crisis [75, 76]. Other actions underpinned by dynamic capabilities such as introducing new products and methods of selling are less important, only appearing in configurations where businesses had not increased operations and had used the furlough scheme. In these configurations, firms tended to perform worse or were able to maintain the same level of performance but were not able to enhance performance by drawing on dynamic capabilities. Configurations also suggest that dynamic capabilities and government support may not be complementary in enhancing performance during a crisis, with government support not featuring as an important factor for firms who enhanced performance by drawing on dynamic capabilities to increase operations.

SMEs aiming to increase performance during a crisis should focus on utilising dynamic capabilities to pivot, increasing operations. This means SMEs should focus on developing dynamic capabilities outside of crisis times, to ensure they are prepared and ready to deploy these capabilities in response to external turbulence. SMEs lacking the capability to respond to a crisis may rely on government support which may help the firm to survive, but at best this is likely to result in maintaining performance, with an increased likelihood of decreased performance.

In summary, the study contributes to the literature by identifying configurations of factors that are related to the performance of SMEs during the pandemic, highlighting non-linear and asymmetric relationships, and by identifying dominant predictors. These contributions extend the existing literature, which has focused on symmetric and linear techniques such as regression. From a methodological perspective, the study highlights how management researchers can apply tree based machine learning approaches, alongside XAI techniques to derive novel insights about existing management problems.

6 Conclusions

The empirical evidence for complexity, enabled by algorithm-supported induction, forms a basis for guiding future research directions, hypotheses, and strategies for SMEs to navigate crises and enhance resilience. This study identifies various configurations and heterogeneity in the strategic responses to COVID-19 and their impact on performance, as well as asymmetric and non-linear relationships.

Within the labyrinth of the COVID-19 pandemic's repercussions on the business milieu, this study presents a discerning exploration of the strategic responses of SMEs, shedding light on the interplay between their capabilities, governmental aid, and contextual variables. Situating this within the contours of complexity theory, the insights presented in this study unravel how SMEs navigate, adapt, and respond to challenges, engendering variegated performance outcomes.

The research carried out in this study delineates three salient strategic trajectories adopted by SMEs in their navigation of the pandemic's complexities. These trajectories are anchored in a diverse blend of strategic actions and contextual factors, and their association with different levels of performance, which not only reinforce extant literature on strategic responses to the pandemic [20, 67], but also extend its horizons by employing a machine learning configurational approach. This methodology, particularly the dominance analysis approach, is a novel contribution, spotlighting the primacy of SMEs adapting to lockdowns and utilising the furlough scheme.

However, these actions are associated with different strategic responses, which are interpreted as pivoting for firms that increase operations and retrenchment for firms that cope by temporarily reducing staff. These actions are also associated with different performance outcomes, with the former related to improved performance, and the latter related to decreased performance. The imperative here is the strategic sagacity required by SMEs in crises, recognizing that action-reaction dynamics, as depicted in complexity theory, might yield contrasting performance outcomes.

From a practical perspective, this suggests that SMEs need to carefully consider their strategic responses to crises since different combinations of factors can have various performance outcomes. Therefore, SMEs should move away from retrenchment towards ensuring they have the capability to pivot if necessary. Rather than relying solely on cost-cutting measures, firms should proactively invest in strategies that enable them to seize emerging opportunities and maintain long-term competitiveness. At the same time, policymakers should recognise that support mechanisms that facilitate retrenchment may lead to at least temporary decreases in performance. However, these firms have not been forced to close, which is a positive outcome.

Additionally, this study makes methodological contributions by adopting a machine learning approach to examine configurations. Few studies have employed machine learning to explore configurations, and these studies have primarily focused on other contexts [33, 45, 148]. In addition to revealing configurations, tree based approaches have advantages over traditional regression models which have stringent assumptions such as absence of multicollinearity and underlying linear relationships, as well as a focus on net effects [69, 113]. The decision trees are further supplemented with permutation variable importance, contributing to the wider dominance analysis literature [41, 60, 64] and along with the PDPs to the small body of management literature focusing on explainable AI [149, 150].

This understanding, enabled by algorithm supported induction, underscores the necessity for future research directions, hypotheses, and strategies to be informed by the complexity inherent in SMEs' responses to crises. By doing so, this study advances the fields of crisis management and SME research, providing an empirical basis for addressing the multifaceted challenges faced by SMEs in times of crisis.

This study is not without limitations. The sample for the study is limited to UK SMEs, which may limit the generalisability of the results to larger firms or firms in other countries. Although global, the effect of the COVID-19 pandemic, and government responses, differed across countries. Future work could therefore consider applying a similar machine learning method to study the impact of SMEs crisis responses alongside government support mechanisms in other countries, or could compare the effect of different government responses across countries. Future work could also consider the strategic responses of larger firms.

Although the focus is on interpretation rather than prediction, both models display a modest level of predictive accuracy, reflecting the complexity and variability in SME decision making processes. The focus is also on short term performance measures, and in particular on revenue, which is only one dimension of performance during a crisis. Future work can consider other dimensions of performance alongside medium and long term outcomes from responses to the pandemic. The focus is also placed on specific response mechanisms and contextual factors, rather than directly measuring broader strategic responses. Although this enables us to link back to the wider strategic management literature [20] by inferring broader strategies, future studies could consider expanding measurement instruments to more directly assess overarching strategic responses which may govern individual mechanisms, or could consider a wider range of resources and capabilities, such as the role of digital capabilities.

The adoption of machine learning approaches in the management literature is also relatively new, and it is anticipated that these techniques will evolve as they become more popular. Building on the algorithm supported induction approach used in the present study, future work could also consider focusing on formally testing some of the interactions observed in this study, as well as predicting future performance outcomes.

Appendix

See appendix Table 2.

Table 2 Model Accuracy metrics calculated by making predictions using the test dataset

Metric	Decision Tree	GBM
Multiclass AUC-ROC	0.61	0.66
Kappa	0.18	0.19
Sensitivity		
Class: up	0.58	0.62
Class: down	0.62	0.62
Class: same	0.35	0.41
Specificity		
Class: up	0.85	0.85
Class: down	0.75	0.78
Class: same	0.75	0.76
Balanced Accuracy		
Class: up	0.71	0.73
Class: down	0.69	0.70
Class: same	0.55	0.58

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Declarations

Conflicts of interests The authors have no relevant financial or non-financial interests to disclose.

References

1. Brown, R., & Rocha, A. (2020). Entrepreneurial uncertainty during the Covid-19 crisis: Mapping the temporal dynamics of entrepreneurial finance. *Journal of Business Venturing Insights, 14*, e00174. https://doi.org/10.1016/j.jbvi.2020.e00174
2. Krammer, S. (2022). Navigating the new normal: which firms have adapted better to the COVID-19 disruption? *Technovation, 110*, 102368. https://doi.org/10.1016/j.technovation.2021.102368
3. Stephens, M., Cross, S., Luckwell, G. (2020). Coronavirus and the impact on output in the UK economy. Office for National Statistics: 1–29.
4. Brown, R., & Cowling, M. (2021). The geographical impact of the Covid-19 crisis on precautionary savings, firm survival and jobs: Evidence from the United Kingdom's 100 largest towns and cities. *International Small Business Journal: Researching Entrepreneurship, 39*, 319–329. https://doi.org/10.1177/0266242621989326

5. Piva, E., & Guerini, M. (2022). The impact of the COVID-19 pandemic and pandemic-related policies on new firm creation: an analysis of the Italian case. *Small Business Economics*. https://doi.org/10.1007/s11187-022-00621-w
6. Bartik, A., Cullen, Z., Bertrand, M., Glaeser, E. L., Luca, M., & Stanton, C. (2020). How are small businesses adjusting to COVID-19? Early evidence from a survey. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.3570896
7. Eggers, F. (2020). Masters of disasters? Challenges and opportunities for SMEs in times of crisis. *Journal of Business Research, 116*, 199–208. https://doi.org/10.1016/j.jbusres.2020.05.025
8. Calabrese, R., Cowling, M., & Liu, W. (2022). Understanding the dynamics of UK covid-19 SME financing. *British Journal of Management, 33*, 657–677. https://doi.org/10.1111/1467-8551.12576
9. Cowling, M., Brown, R., & Rocha, A. (2020). Did you save some cash for a rainy COVID-19 day? The crisis and SMEs. *International Small Business Journal: Researching Entrepreneurship, 38*, 593–604. https://doi.org/10.1177/0266242620945102
10. Belitski, M., Guenther, C., Kritikos, A. S., & Thurik, R. (2022). Economic effects of the COVID-19 pandemic on entrepreneurship and small businesses. *Small Business Economics, 58*, 593–609. https://doi.org/10.1007/s11187-021-00544-y
11. Cruz-Cárdenas, J., Zabelina, E., Guadalupe-Lanas, J., Palacio-Fierro, A., & Ramos-Galarza, C. (2021). COVID-19, consumer behavior, technology, and society: A literature review and bibliometric analysis. *Technological Forecasting and Social Change*. https://doi.org/10.1016/j.techfore.2021.121179
12. Voorhees, C. M., Fombelle, P. W., & Bone, S. A. (2020). Don't forget about the frontline employee during the COVID-19 pandemic: preliminary insights and a research agenda on market shocks. *Journal of Service Research, 23*, 396–400.
13. Guckenbiehl, P., & Corral de Zubielqui, G. (2022). Start-ups' business model changes during the COVID-19 pandemic: Counteracting adversities and pursuing opportunities. *International Small Business Journal., 40*(2), 150–177.
14. Zahoor, N., Golgeci, I., Haapanen, L., Ali, I., & Arslan, A. (2022). The role of dynamic capabilities and strategic agility of B2B high-tech small and medium-sized enterprises during COVID-19 pandemic: Exploratory case studies from Finland. *Industrial Marketing Management, 105*, 502–514. https://doi.org/10.1016/j.indmarman.2022.07.006
15. Maria del Rio-Chanona, R., Mealy, P., Pichler, A., Lafond, F., & Doyne Farmer, J. (2020). Supply and demand shocks in the COVID-19 pandemic: An industry and occupation perspective. *Oxford Review of Economic Policy, 36*, S94–S137. https://doi.org/10.1093/oxrep/graa033
16. Morgan, T., Anokhin, S., Ofstein, L., & Friske, W. (2020). SME response to major exogenous shocks : The bright and dark sides of business model pivoting. *International Small Business Journal, 38*, 369–379. https://doi.org/10.1177/0266242620936590
17. Higueras-castillo, E., Li, F. J., & Villarejo-ramos, F. (2023). Intention to use e-commerce vs physical shopping difference between consumers in the post-COVID era. *Journal of Business Research, 157*, 113622. https://doi.org/10.1016/j.jbusres.2022.113622
18. Manolova, T. S., Brush, C. G., Edelman, L. F., & Elam, A. (2020). Pivoting to stay the course: How women entrepreneurs take advantage of opportunities created by the COVID-19 pandemic. *International Small Business Journal: Researching Entrepreneurship, 38*, 481–491. https://doi.org/10.1177/0266242620949136
19. Conti, C. R., Goldszmidt, R., & de Vasconcelos, F. C. (2020). Firm characteristics and capabilities that enable superior performance in recessions. *Journal of business research., 1*(119), 553–561.
20. Wenzel, M., Stanske, S., & Lieberman, M. B. (2020). Strategic responses to crisis. *Strategic Management Journal, 41*, 3161. https://doi.org/10.1002/smj.3161
21. Clampit, J. A., Lorenz, M. P., Gamble, J. E., & Lee, J. (2021). Performance stability among small and medium-sized enterprises during COVID-19: A test of the efficacy of dynamic capabilities. *International Small Business Journal: Researching Entrepreneurship, 40*, 403–419. https://doi.org/10.1177/02662426211033270
22. Mukherjee, A., Scott, J. M., Deakins, D., & McGlade, P. (2023). "Stay home, save SMEs"? The impact of a unique strict COVID-19 lockdown on small businesses. *International Journal of Entrepreneurial Behavior and Research, 29*, 1884–1905. https://doi.org/10.1108/IJEBR-02-2023-0099
23. Balta, M. E., Papadopoulos, T., & Spanaki, K. (2023). Business model pivoting and digital technologies in turbulent environments. *International Journal of Entrepreneurial Behavior and Research*. https://doi.org/10.1108/IJEBR-02-2023-0210

24. Dejardin, M., Raposo, M. L., Ferreira, J. J., Fernandes, C. I., Veiga, P. M., & Farinha, L. (2022). The impact of dynamic capabilities on SME performance during COVID-19. *Review of Managerial Science, 2018*, 11482. https://doi.org/10.1007/s11846-022-00569-x
25. Matikonis, K., & Graham, B. (2024). Dynamic capabilities and employment during Covid-19: The moderating effect of government support. *International Small Business Journal, 42*, 157–184. https://doi.org/10.1177/02662426231173352
26. Belghitar, Y., Moro, A., & Radić, N. (2022). When the rainy day is the worst hurricane ever: the effects of governmental policies on SMEs during COVID-19. *Small Business Economics, 58*, 943–961. https://doi.org/10.1007/s11187-021-00510-8
27. Jibril, H., Roper, S., Hart, M. (2021). COVID-19, business support and SME productivity in the UK. The Productivity Institute.
28. Cowling, M., Wilson, N., Nightingale, P., & Kacer, M. (2022). Predicting future default on the Covid-19 bounce back loan scheme: The £46.5 billion question. *International Small Business Journal: Researching Entrepreneurship, 40*, 650–666. https://doi.org/10.1177/02662426211058050
29. Dörr, J. O., Licht, G., & Murmann, S. (2022). Small firms and the COVID-19 insolvency gap. *Small Business Economics, 58*, 887–917. https://doi.org/10.1007/S11187-021-00514-4/TABLES/12
30. Klyver, K., & Nielsen, S. L. (2021). Which crisis strategies are (expectedly) effective among SMEs during COVID-19? *Journal of Business Venturing Insights, 16*, e00273. https://doi.org/10.1016/j.jbvi.2021.e00273
31. Eisenhardt, K. M., & Martin, J. A. (2000). Dynamic capabilities: What are they? *Strategic Management Journal, 21*, 1105–1121. https://doi.org/10.1002/1097-0266(200010/11)21:10/11%3c1105::AID-SMJ133%3e3.0.CO;2-E
32. Matikonis, K., & Gobey, M. (2024). Small business property tax reductions and firm productivity. *Small Business Economics, 62*, 307–324. https://doi.org/10.1007/s11187-023-00768-0
33. Tidhar, R., & Eisenhardt, K. M. (2020). Get rich or die trying… finding revenue model fit using machine learning and multiple cases. *Strategic Management Journal, 41*, 1245–1273. https://doi.org/10.1002/smj.3142
34. Vergne, J. P., & Depeyre, C. (2016). How do firms adapt? a fuzzy-set analysis of the role of cognition and capabilities in u.s. defense firms' responses to 9/11. *Academy of Management Journal, 59*, 1653–1680. https://doi.org/10.5465/amj.2013.1222
35. Woodside, A. G. (2014). Embrace perform model: Complexity theory, contrarian case analysis, and multiple realities. *Journal of Business Research, 67*, 2495–2503. https://doi.org/10.1016/j.jbusres.2014.07.006
36. Newbert, S. L., Kher, R., & Yang, S. (2022). Now that's interesting and important! Moving beyond averages to increase the inferential value of empirical findings in entrepreneurship research. *Journal of Business Venturing, 37*, 106185. https://doi.org/10.1016/j.jbusvent.2021.106185
37. Sharma, G. D., Kraus, S., Talan, A., Srivastava, M., & Theodoraki, C. (2024). Navigating the storm: the SME way of tackling the pandemic crisis. *Small Business Economics, 63*, 221–241. https://doi.org/10.1007/s11187-023-00810-1
38. Kleine, A., Rebecca, B. G., & De Massis, A. (2023). Look in to look out: Strategy and family business identity during COVID-19. *Small Business Economics*. https://doi.org/10.1007/s11187-023-00846-3
39. Liu, C. Y., & Nazareno, L. (2024). State responses during the COVID-19 pandemic and their impacts on small businesses. *Small Business Economics*. https://doi.org/10.1007/s11187-024-00923-1
40. Oikawa, M., & Onishi, K. (2024). Impact of financial support expansion on restaurant entries and exits during the COVID-19 pandemic. *Small Business Economics*. https://doi.org/10.1007/s11187-024-00912-4
41. Azen, R., & Budescu, D. (2003). The dominance analysis approach for comparing predictors in multiple regression. *Psychological Methods, 8*, 129–148. https://doi.org/10.1037/1082-989X.8.2.129
42. Hamann, R., Sewlal, A., Pariag-Maraye, N., Muthuri, J., Amaeshi, K., Nwagwu, I., & Soderbergh, J. (2022). Strengthening or Restricting? Explaining the Covid-19 Pandemic's configurational effects on companies' sustainability strategies and practices. *Business and Society*. https://doi.org/10.1177/00076503221134100

43. Puumalainen, K., Sjögrén, H., Soininen, J., Syrjä, P., & Kraus, S. (2023). Crisis response strategies and entrepreneurial orientation of SMEs: A configurational analysis on performance impacts. *International Entrepreneurship and Management Journal, 19*(4), 1527–1559. https://doi.org/10.1007/s11365-023-00847-4
44. Kusa, R., Duda, J., & Suder, M. (2022). How to sustain company growth in times of crisis: The mitigating role of entrepreneurial management. *Journal of Business Research, 142*, 377–386. https://doi.org/10.1016/j.jbusres.2021.12.081
45. Graham, B., & Bonner, K. (2022). One size fits all? Using machine learning to study heterogeneity and dominance in the determinants of early stage entrepreneurship. *Journal of Business Research, 152*, 42–59. https://doi.org/10.2139/ssrn.3997841
46. Graham, B., & Bonner, K. (2024). The role of institutions in early-stage entrepreneurship: An explainable artificial intelligence approach. *Journal of Business Research, 175*, 114567. https://doi.org/10.1016/j.jbusres.2024.114567
47. Delen, D., & Zolbanin, H. M. (2018). The analytics paradigm in business research. *Journal of Business Research, 90*, 186–195. https://doi.org/10.1016/j.jbusres.2018.05.013
48. Shrestha, Y. R., He, V. F., Puranam, P., & von Krogh, G. (2021). Algorithm supported induction for building theory: How can we use prediction models to theorize? *Organization Science, 32*, 856–880. https://doi.org/10.1287/orsc.2020.1382
49. Chae, H. C. (2023). In search of gazelles : machine learning prediction for Korean high—growth firms. *Small Business Economics, 62,* 243–284. https://doi.org/10.1007/s11187-023-00760-8
50. Ballestar, M. T., Grau-Carles, P., & Sainz, J. (2019). Predicting customer quality in e-commerce social networks: a machine learning approach. *Review of Managerial Science, 13*, 589–603. https://doi.org/10.1007/s11846-018-0316-x
51. Lévesque, M., Obschonka, M., & Nambisan, S. (2020). Pursuing impactful entrepreneurship research using artificial intelligence. *Entrepreneurship Theory and Practice, 46*, 803–832.
52. Obschonka, M., & Audretsch, D. B. (2019). Artificial intelligence and big data in entrepreneurship: a new era has begun. *Small Business Economics, 55*, 529–539. https://doi.org/10.1007/s11187-019-00202-4
53. Razzak, B. M., Idris, B., Hasan, R., Saridakis, G., & Hansen, J. M. (2023). The impact of Covid-19 on struggling ethnic minority entrepreneurs' business strategy: The case of Bangladeshi curry houses in the United Kingdom. *International Journal of Entrepreneurial Behavior and Research, 29*, 1837–1866. https://doi.org/10.1108/IJEBR-05-2021-0389
54. Haneberg, D. H. (2021). How combinations of network participation, firm age and firm size explain SMEs' responses to COVID-19. *Small Enterprise Research, 28*, 229–246. https://doi.org/10.1080/13215906.2021.1989626
55. Wang, Y., Turkina, E., Khoury, S., & Lemay, N. (2024). Causal configurations of SME strategic renewal in crisis: qualitative comparative analysis (QCA) of Quebec entrepreneurs amid COVID-19. *Entrepreneurship and Regional Development, 36*, 745–774. https://doi.org/10.1080/08985626.2023.2223158
56. Drăgan, G. B., Arfi, W. B., Tiberius, V., & Ammari, A. (2023). Gravitating exogenous shocks to the next normal through entrepreneurial coopetive interactions: A PLS-SEM and fsQCA approach. *Journal of Business Research*. https://doi.org/10.1016/j.jbusres.2022.113627
57. Jabeur, S. B., Gharib, C., Mefteh-Wali, S., & Arfi, W. B. (2021). CatBoost model and artificial intelligence techniques for corporate failure prediction. *Technological Forecasting and Social Change, 166*, 120658. https://doi.org/10.1016/j.techfore.2021.120658
58. Żbikowski, K., & Antosiuk, P. (2021). A machine learning, bias-free approach for predicting business success using Crunchbase data. *Information Processing and Management, 58*, 102555. https://doi.org/10.1016/j.ipm.2021.102555
59. Jabeur, S. B., Ballouk, H., Mefteh-Wali, S., & Omri, A. (2021). Forecasting the macrolevel determinants of entrepreneurial opportunities using artificial intelligence models. *Technological Forecasting and Social Change, 175*, 121353. https://doi.org/10.1016/j.techfore.2021.121353
60. Arin, P., Huang, V. Z., Minniti, M., Nandialath, A. M., & Reich, O. F. M. (2015). Revisiting the determinants of entrepreneurship: A Bayesian approach. *Journal of Management, 41*, 607–631. https://doi.org/10.1177/0149206314558488
61. Connelly, B. L., Russell Crook, T., Combs, J. G., Ketchen, D. J., & Aguinis, H. (2018). Competence- and integrity-based trust in Interorganizational relationships: Which matters more? *Journal of Management, 44*, 919–945. https://doi.org/10.1177/0149206315596813

62. Bansal, H. S., & Duverger, P. (2013). Investigating the measures of relative importance in marketing research. *International Journal of Market Research, 55*, 675–694. https://doi.org/10.2501/IJMR-2013-057
63. Viglia, G., & Abrate, G. (2017). When distinction does not pay off—Investigating the determinants of European agritourism prices. *Journal of Business Research, 80*, 45–52. https://doi.org/10.1016/j.jbusres.2017.07.004
64. Budescu, D. V., & Azen, R. (2004). Beyond global measures of relative importance: Some insights from dominance analysis. *Organizational Research Methods, 7*, 341–350. https://doi.org/10.1177/1094428104267049
65. Azen, R., & Budescu, D. V. (2006). Comparing predictors in multivariate regression models: An extension of dominance analysis. *Journal of Educational Research Association and American Statistical Association, 31*, 157–180.
66. Basu, Shubhabrata, Surender Munjal, Pawan Budhwar, and Pradeep Misra. 2022. Entrepreneurial Adaptation in Emerging Markets: Strategic Entrepreneurial Choices, Adaptive Capabilities and Firm Performance. *British Journal of Management* 33. John Wiley & Sons, Ltd: 1864–1886. https://doi.org/10.1111/1467-8551.12572.
67. Puthusserry, P., King, T., Miller, K., & Khan, Z. (2022). A Typology of emerging market SMEs' COVID-19 response strategies: The role of TMTs and organizational design. *British Journal of Management, 33*, 603–633. https://doi.org/10.1111/1467-8551.12591
68. Wulandhari, N. B., Ingga, P. B., Mishra, N., Akbar, S., Do, Q., & Milligan, G. (2022). Organizational resilience to supply chain risks during the COVID-19 pandemic. *British Journal of Management*. https://doi.org/10.1111/1467-8551.12648
69. Woodside, A. G. (2013). Moving beyond multiple regression analysis to algorithms: Calling for adoption of a paradigm shift from symmetric to asymmetric thinking in data analysis and crafting theory. *Journal of Business Research, 66*, 463–472. https://doi.org/10.1016/j.jbusres.2012.12.021
70. Koch, J., & Schermuly, C. C. (2021). Managing the crisis: How COVID-19 demands interact with agile project management in predicting employee exhaustion. *British Journal of Management, 32*, 1265–1283. https://doi.org/10.1111/1467-8551.12536
71. Morgan, T., Anokhin, S., Ofstein, L., & Friske, W. (2020). SME response to major exogenous shocks: The bright and dark sides of business model pivoting. *International Small Business Journal: Researching Entrepreneurship, 38*, 369–379. https://doi.org/10.1177/0266242620936590
72. Clauss, T., Breier, M., Kraus, S., Durst, S., & Mahto, R. V. (2021). Temporary business model innovation—SMEs' innovation response to the Covid-19 crisis. *R&D Management, 52*, 295–312. https://doi.org/10.1111/radm.12498
73. Lefebvre, V. (2023). Mobilizing potential slack and firm performance: evidence from French SMEs before and during the COVID-19 period. *Journal of Small Business Management*. https://doi.org/10.1080/00472778.2023.2182440
74. Hausman, A., & Johnston, W. J. (2014). The role of innovation in driving the economy: Lessons from the global financial crisis. *Journal of Business Research, 67*, 2720–2726. https://doi.org/10.1016/j.jbusres.2013.03.021
75. Teece, D., Pisano, G., & Shuen, A. (1997). Dynamic capabilities and strategic management. *Strategic Management Journal, 18*, 509–533. https://doi.org/10.1002/(SICI)1097-0266(199708)18:7%3c509::AID-SMJ882%3e3.0.CO;2-Z
76. Helfat, C., & Peteraf, M. (2009). Understanding dynamic capabilities: Progress along a developmental path. *Strategic Organization, 7*, 91–102. https://doi.org/10.1177/1476127008100133
77. Winter, S. G. (2003). Understanding dynamic capabilities. *Strategic Management Journal, 24*, 991–995. https://doi.org/10.1002/smj.318
78. Bingham, C. B., Heimeriks, K. H., Schijven, M., & Gates, S. (2015). Concurrent learning: How firms develop multiple dynamic capabilities in parallel. *Strategic Management Journal, 36*, 1802–1825. https://doi.org/10.1002/smj.2347
79. Arend, R. J. (2014). Entrepreneurship and dynamic capabilities: How firm age and size affect the "capability enhancement-SME performance" relationship. *Small Business Economics, 42*, 33–57. https://doi.org/10.1007/s11187-012-9461-9
80. Uhlaner, L. M., van Stel, A., Duplat, V., & Zhou, H. (2013). Disentangling the effects of organizational capabilities, innovation and firm size on SME sales growth. *Small Business Economics, 41*, 581–607. https://doi.org/10.1007/s11187-012-9455-7

81. Battisti, M., & Deakins, D. (2017). The relationship between dynamic capabilities, the firm's resource base and performance in a post-disaster environment. *International Small Business Journal: Researching Entrepreneurship, 35*, 78–98. https://doi.org/10.1177/0266242615611471
82. Jantunen, A., Ellonen, H. K., & Johansson, A. (2012). Beyond appearances—do dynamic capabilities of innovative firms actually differ? *European Management Journal, 30*, 141–155. https://doi.org/10.1016/j.emj.2011.10.005
83. Mahto, R. V., Llanos-Contreras, O., & Hebles, M. (2022). Post-disaster recovery for family firms: The role of owner motivations, firm resources, and dynamic capabilities. *Journal of Business Research, 145*, 117–129. https://doi.org/10.1016/j.jbusres.2022.02.089
84. Seetharaman, P. (2020). Business models shifts: Impact of Covid-19. *International Journal of Information Management, 54*, 102173. https://doi.org/10.1016/j.ijinfomgt.2020.102173
85. Breier, M., Kallmuenzer, A., Clauss, T., Gast, J., Kraus, S., & Tiberius, V. (2021). The role of business model innovation in the hospitality industry during the COVID-19 crisis. *International Journal of Hospitality Management, 92*, 102723. https://doi.org/10.1016/j.ijhm.2020.102723
86. Björklund, T. A., Mikkonen, M., Mattila, P., & van der Marel, F. (2020). Expanding entrepreneurial solution spaces in times of crisis: Business model experimentation amongst packaged food and beverage ventures. *Journal of Business Venturing Insights, 14*, e00197. https://doi.org/10.1016/j.jbvi.2020.e00197
87. Dyduch, W., Chudziński, P., Cyfert, S., & Zastempowski, M. (2021). Dynamic capabilities, value creation and value capture: Evidence from SMEs under Covid-19 lockdown in Poland. *PLoS ONE, 16*, e0252423. https://doi.org/10.1371/journal.pone.0252423
88. Ozanne, L. K., Chowdhury, M., Prayag, G., & Mollenkopf, D. A. (2022). SMEs navigating COVID-19: The influence of social capital and dynamic capabilities on organizational resilience. *Industrial Marketing Management, 104*, 116–135. https://doi.org/10.1016/j.indmarman.2022.04.009
89. Barreto, I. (2010). Dynamic capabilities: a review of past research and an agenda for the future. *Journal of Management, 36*, 256–280. https://doi.org/10.1177/0149206309350776
90. Baia, E., Ferreira, J. J., & Rodrigues, R. (2020). Value and rareness of resources and capabilities as sources of competitive advantage and superior performance. *Knowledge Management Research and Practice, 18*, 249–262. https://doi.org/10.1080/14778238.2019.1599308
91. Wilden, R., Devinney, T. M., & Dowling, G. R. (2016). The architecture of dynamic capability research identifying the building blocks of a configurational approach. *Academy of Management Annals, 10*, 997–1076. https://doi.org/10.1080/19416520.2016.1161966
92. Zahra, S. A., Sapienza, H. J., & Davidsson, P. (2006). Entrepreneurship and dynamic capabilities: A review, model and research agenda. *Journal of Management Studies, 43*, 917–955. https://doi.org/10.1111/j.1467-6486.2006.00616.x
93. Helfat, C., Finkelstein, S., Mitchell, W., Peteraf, M., Singh, H., Teece, D., Winter, S. (2007). *Dynamic capabilities : understanding strategic change in organizations*. Malden: Blackwell Pub.
94. Chakrabarti, A. (2015). Organizational adaptation in an economic shock: The role of growth reconfiguration. *Strategic Management Journal, 36*, 1717–1738. https://doi.org/10.1002/smj
95. Miklian, J., & Hoelscher, K. (2022). SMEs and exogenous shocks: A conceptual literature review and forward research agenda. *International Small Business Journal: Researching Entrepreneurship, 40*, 178–204. https://doi.org/10.1177/02662426211050796
96. HM Treasury. (2020). Chancellor of the Exchequer, Rishi Sunak on COVID19 response.
97. Granja, J., Makridis, C., Yannelis, C., & Zwick, E. (2022). Did the paycheck protection program hit the target? *Journal of Financial Economics, 145*, 725–761. https://doi.org/10.1016/j.jfineco.2022.05.006
98. Davidsson, P., Recker, J., & von Briel, F. (2021). COVID-19 as External Enabler of entrepreneurship practice and research. *BRQ Business Research Quarterly, 24*, 214–223. https://doi.org/10.1177/23409444211008902
99. Adalet McGowan, M., Andrews, D., & Millot, V. (2018). The walking dead? Zombie firms and productivity performance in OECD countries. *Economic Policy, 33*(96), 685–736.
100. Acharya, V. V., Crosignani, M., Eisert, T., & Steffen, S. (2022). Zombie lending: theoretical, international, and historical perspectives. *Annual Review of Financial Economics, 14*, 21–38. https://doi.org/10.1146/annurev-financial-111620-114424
101. Hoshi, T., Kawaguchi, D., & Ueda, K. (2023). Zombies, again? The COVID-19 business support programs in Japan. *Journal of Banking and Finance, 147*, 106421. https://doi.org/10.1016/j.jbankfin.2022.106421

102. Ndofor, H. A., Vanevenhoven, J., & Barker, V. (2013). Software firm turnarounds in the 1990s: An analysis of reversing decline in a growing, dynamic industry. *Strategic Management Journal, 34*, 1123–1133. https://doi.org/10.1002/smj
103. Robbins, K., & Pearce, J. A. (1992). Turnaround: Retrenchment and Recovery. *Strategic Management Journal, 13*, 287–309.
104. de Figueiredo Jr, R. J., Feldman, E. R., & Rawley, E. (2019). The costs of refocusing: Evidence from hedge fund closures during the financial crisis. *Strategic Management Journal, 40*(8), 1268–1290.
105. Bruton, G. D., Ahlstrom, D., & Wan, J. C. C. (2003). Turnaround in East Asian firms: Evidence from ethnic overseas Chinese communities. *Strategic Management Journal, 24*, 519–540. https://doi.org/10.1002/smj.312
106. Thorgren, S., & Alma, T. (2020). Staying alive during an unfolding crisis : How SMEs ward off impending disaster. *Journal of Business Venturing Insights, 14*, e00187. https://doi.org/10.1016/j.jbvi.2020.e00187
107. Office for National Statistics. (2022). How furlough and changes in the employee workforce have affected earnings growth during the coronavirus (COVID-19) pandemic, UK : 2020 to 2021.
108. Francis-Devine, B., Powell, A., Clark, H. (2021). Coronavirus Job Retention Scheme: statistics. House of Commons Library Briefing Paper.
109. Awe, O. A., Woodside, A. G., Nerur, S., & Prater, E. (2020). Capturing heterogeneities in orchestrating resources for accurately forecasting high (separately low) project management performance. *International Journal of Production Economics, 224*, 107556. https://doi.org/10.1016/j.ijpe.2019.107556
110. Arora, S. K., Li, Y., Youtie, J., & Shapira, P. (2020). Measuring dynamic capabilities in new ventures: exploring strategic change in US green goods manufacturing using website data. *Journal of Technology Transfer, 45*, 1451–1480. https://doi.org/10.1007/s10961-019-09751-y
111. Wilden, R., Gudergan, S., Akaka, M. A., Averdung, A., & Teichert, T. (2019). The role of cocreation and dynamic capabilities in service provision and performance: A configurational study. *Industrial Marketing Management, 78*, 43–57. https://doi.org/10.1016/j.indmarman.2018.06.008
112. Meyer, A., Tsui, A., & Hinings, C. (1993). Configurational approaches to organizational analysis. *Academy of Management Journal, 36*, 1175–1195.
113. Fiss, P. C. (2007). A set-theoretic approach to organizational configurations. *Academy of Management Review, 32*, 1180–1198.
114. Liu, Y., Mezei, J., Kostakos, V., & Li, H. (2017). Applying configurational analysis to IS behavioural research: A methodological alternative for modelling combinatorial complexities. *Information Systems Journal, 27*, 59–89. https://doi.org/10.1111/isj.12094
115. Beynon, M. J., Battisti, M., Jones, P., & Pickernell, D. (2021). How institutions matter in the context of business exit: A country comparison using GEM data and fsQCA. *British Journal of Management, 32*, 832–851. https://doi.org/10.1111/1467-8551.12438
116. Gelhard, C., von Delft, S., & Gudergan, S. P. (2016). Heterogeneity in dynamic capability configurations: Equifinality and strategic performance. *Journal of Business Research, 69*, 5272–5279. https://doi.org/10.1016/j.jbusres.2016.04.124
117. Jantunen, A., Tarkiainen, A., Chari, S., & Oghazi, P. (2018). Dynamic capabilities, operational changes, and performance outcomes in the media industry. *Journal of Business Research, 89*, 251–257. https://doi.org/10.1016/j.jbusres.2018.01.037
118. Schade, P., & Schuhmacher, M. C. (2023). Predicting entrepreneurial activity using machine learning. *Journal of Business Venturing Insights, 19*, e00357. https://doi.org/10.1016/j.jbvi.2022.e00357
119. Wang, H., Zhao, T., Cooper, S. Y., Wang, S., Harrison, R. T., & Yang, Z. (2023). Effective policy mixes in entrepreneurial ecosystems: a configurational analysis in China. *Small Business Economics, 60*, 1509–1542. https://doi.org/10.1007/s11187-022-00658-x
120. Xie, Z., Wang, X., Xie, L., Dun, S., & Li, J. (2021). Institutional context and female entrepreneurship: A country-based comparison using fsQCA. *Journal of Business Research, 132*, 470–480. https://doi.org/10.1016/j.jbusres.2021.04.045
121. Jiao, J. L., Zhang, X. L., & Tang, Y. S. (2020). What factors determine the survival of green innovative enterprises in China?–A method based on fsQCA. *Technology in Society., 1*(62), 101314.

122. Malik, O. R., & Kotabe, M. (2009). Dynamic capabilities, government policies, and performance in firms from emerging economies: Evidence from India and Pakistan. *Journal of Management Studies, 46*, 421–450. https://doi.org/10.1111/j.1467-6486.2008.00817.x
123. Choudhury, P., Allen, R. T., & Endres, M. G. (2021). Machine learning for pattern discovery in management research. *Strategic Management Journal, 42*, 30–57. https://doi.org/10.1002/smj.3215
124. Yarkoni, T., & Westfall, J. (2017). Choosing prediction over explanation in psychology: lessons from machine learning. *Perspectives on Psychological Science, 12*, 1100–1122. https://doi.org/10.1177/1745691617693393
125. Sheetal, A., & Jiang, Z. (2022). Using machine learning to analyze longitudinal data : A tutorial guide and best-practice recommendations for social science researchers. *Applied Psychology*. https://doi.org/10.1111/apps.12435
126. Chou, Y. C., Chuang, H. H. C., Chou, P., & Oliva, R. (2023). Supervised machine learning for theory building and testing: Opportunities in operations management. *Journal of Operations Management*. https://doi.org/10.1002/joom.1228
127. von Krogh, G. (2020). Building capacity for empirical discovering in management and organization studies. *Academy of Management Discoveries, 6*, 159–164. https://doi.org/10.5465/amd.2020.0053
128. Department for Business Energy and Industrial Strategy. (2022). Longitudinal small business survey, 2015–2020. UK Data Service. https://doi.org/10.5255/UKDA-SN-7973-6
129. Kump, B., Engelmann, A., Kessler, A., & Schweiger, C. (2019). Toward a dynamic capabilities scale: Measuring organizational sensing, seizing, and transforming capacities. *Industrial and Corporate Change, 28*, 1149–1172. https://doi.org/10.1093/icc/dty054
130. Pavlou, P. A., & El Sawy, O. A. (2010). The "third hand": IT-enabled competitive advantage in turbulence through improvisational capabilities. *Information Systems Research, 21*, 443–471. https://doi.org/10.1287/isre.1100.0280
131. Cesaroni, F. M., Sentuti, A., & Buratti, A. (2015). Same Crisis diffeernt strategies? Italian men and women entrepreneurs in front of the conomic recession. *Journal of research in gender studies, 5*, 205–231.
132. Li, F., Wang, L., Jin, Z., Huang, L., & Xia, B. (2020). Key factors affecting sustained business operations after an earthquake: a case study from New Beichuan, China, 2013–2017. *Natural Hazards, 104*, 101–121. https://doi.org/10.1007/s11069-019-03597-1
133. Clemente-Almendros, J. A., González-González, I., Cerdá-Suárez, L. M., & Seguí-Amortegui, L. A. (2023). Understanding the relevance of family business, gender and value chains for SMEs' innovation in the context of COVID-19. *International Journal of Entrepreneurial Behavior and Research*. https://doi.org/10.1108/IJEBR-12-2021-1028
134. Kuhn, M., & Johnson, K. (2013). *Applied predictive modelling*. Springer.
135. Climent, F., Momparler, A., & Carmona, P. (2019). Anticipating bank distress in the Eurozone: An extreme gradient boosting approach. *Journal of Business Research, 101*, 885–896. https://doi.org/10.1016/j.jbusres.2018.11.015
136. Koumbarakis, P., & Volery, T. (2022). Predicting new venture gestation outcomes with machine learning methods. *Journal of Small Business Management*. https://doi.org/10.1080/00472778.2022.2082453
137. Espinoza-Benavides, J., & Díaz, D. (2019). The entrepreneurial profile after failure. *International Journal of Entrepreneurial Behavior and Research, 25*, 1634–1651. https://doi.org/10.1108/IJEBR-04-2018-0242
138. Capatina, A., Cristea, D. S., Micu, A., Micu, A. E., Empoli, G., & Codignola, F. (2023). Exploring causal recipes of startup acceptance into business incubators: A cross-country study. *International Journal of Entrepreneurial Behavior and Research, 29*, 1584–1612. https://doi.org/10.1108/IJEBR-06-2022-0527
139. Strobl, C., Malley, J., & Tutz, G. (2009). An introduction to recursive partitioning: rationale, application, and characteristics of classification and regression trees, bagging, and random forests. *Psychological Methods, 14*, 323–348. https://doi.org/10.1037/a0016973
140. Molnar, Chris. 2022. *Interpretable machine learning*. 2nd ed. Independent.
141. Ali, I., Arslan, A., Chowdhury, M., Khan, Z., & Tarba, S. Y. (2022). Reimagining global food value chains through effective resilience to COVID-19 shocks and similar future events: A dynamic capability perspective. *Journal of Business Research, 141*, 1–12. https://doi.org/10.1016/j.jbusres.2021.12.006

142. Sørensen, J. B., Lasgaard, M., Willert, M. V., & Larsen, F. B. (2021). The relative importance of work-related and non-work-related stressors and perceived social support on global perceived stress in a cross-sectional population-based sample. *BMC Public Health, 21*, 1–14. https://doi.org/10.1186/s12889-021-10594-2
143. Hakanen, J. J., Ropponen, A., Schaufeli, W. B., & De Witte, H. (2019). Who is engaged at work?: A large-scale study in 30 European countries. *Journal of Occupational and Environmental Medicine, 61*, 373–381. https://doi.org/10.1097/JOM.0000000000001528
144. Birhanu, A. G., & Getachew, Y. S. (2022). Gender differences in enterprise performance during the COVID-19 crisis: Do public policy responses matter? *Entrepreneurship Theory and Practice, 46*, 1374–1401. https://doi.org/10.1177/10422587221077222
145. Cowling, M., Liu, W., & Zhang, N. (2018). Did firm age, experience, and access to finance count? SME performance after the global financial crisis. *Journal of Evolutionary Economics, 28*, 77–100. https://doi.org/10.1007/s00191-017-0502-z
146. Simón-Moya, V., Revuelto-Taboada, L., & Ribeiro-Soriano, D. (2016). Influence of economic crisis on new SME survival: reality or fiction? *Entrepreneurship & Regional Development., 28*(1–2), 157–176.
147. Archibugi, D., Filippetti, A., & Frenz, Ma. (2013). The impact of the economic crisis on innovation : Evidence from Europe. *Technological Forecasting & Social Change, 80*, 1247–1260. https://doi.org/10.1016/j.techfore.2013.05.005
148. Duchessi, P., & Lauría, E. J. M. (2013). Decision tree models for profiling ski resorts' promotional and advertising strategies and the impact on sales. *Expert Systems with Applications, 40*, 5822–5829. https://doi.org/10.1016/j.eswa.2013.05.017
149. Zhang, Z., Wu, C., Qu, S., & Chen, X. (2022). An explainable artificial intelligence approach for financial distress prediction. *Information Processing & Management., 59*(4), 102988.
150. Haque, A. B., Islam, A. N., & Mikalef, P. (2023). Explainable Artificial Intelligence (XAI) from a user perspective: A synthesis of prior literature and problematizing avenues for future research. *Technological Forecasting and Social Change., 1*(186), 122120.